复旦数学资料汇编
代数数论进阶

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1.1　试卷: 代数数论进阶

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1. Judgement (Just write true or false)

   (1) K is a number field. Every open subgroup of C_K corresponds to a unique Galois extension of K.

   (2) Local Artin map of a p-adic field is injective but not surjective.

   (3) Let G be a finite group with $|G| = m$. Then for any G-module A, $H^i(G, A)$ is m-torsion for every $i \geq 0$.

   (4) Global Artin map is surjective but not injective.

   (5) Let E/K be a finite extension of number fields, then $\mathrm{Spl}_{E/K}$ is of density $1/[E : K]$.

2. (1) Find all primes p such that there exists a cubic abelian extention of $\mathbb{Q}$ s.t. p is the only ramified prime.

   (2) Classify such cubic abelian extensions in (1) and find their conductors.

3. (1) Compute $H^i(\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p, \mathbb{Z}/p\mathbb{Z})$ for $i = 1, 2$.

   (2) Compute $H^i(\mathbb{Q}_p(\zeta_p), \mathbb{Z}/p\mathbb{Z})$ for $i = 1, 2$.

   (3) Prove that $H^i(\mathbb{Q}_p, \mathbb{Z}/p\mathbb{Z})$ is finite for $i = 1, 2$.

4. (1) Prove that $N_{E/K} : C_E \longrightarrow C_K$ induces $N_{E/K} : \mathrm{Cl}(E) \longrightarrow \mathrm{Cl}(K)$.

   (2) E/K is a finite extension of number fields, let L be the maximal unramified abelian extension in E over K, then $\mathrm{coker}(N_{E/K} : \mathrm{Cl}(E) \to \mathrm{Cl}(K))$ is isomorphic to $\mathrm{Gal}(L/K)$.

   (3) Prove that $|\mathrm{Cl}(\mathbb{Q}(\zeta_n))|$ divides $|\mathrm{Cl}(\mathbb{Q}(\zeta_n + \zeta_n^{-1}))|$.

5. K is a p-adic field containing a primitive n-th root of unity.

   (1) Consider Hilbert symbol: $\langle -, - \rangle_n : K^* \times K^* \longrightarrow \mu_n$, fix $a \in K^*$, $\forall b \in \mathcal{O}_K^*$, $\langle a, b \rangle_n = 1$ iff $K(a^{\frac{1}{n}})$ is unramified over K.

   (2) $K := \mathbb{Q}_3[x]/(x^2 + 1)$, prove K is a field. Compute the residue field of K. Compute K^*/K^{*4}.

   (3) Denote i for $\bar{x} \in K$. Compute $\langle 3, i \rangle_4$, $\langle 3, 1 - i \rangle_4$ and $\langle 3, q \rangle_4$ for prime number q.

6. (1) Let $E = K(\sqrt{a})$, assume $\exists b, c \in \mathcal{O}_K$, s.t. $a = b^2 - 4c$. Consider $\mathfrak{p}$ prime in $\mathcal{O}_K$, s.t., $2 \in \mathfrak{p}$, and $a \notin \mathfrak{p}$. Prove that $\mathfrak{p}$ is unramified in E/K.

   (2) For $K = \mathbb{Q}(\sqrt{-14})$, we have known that $h_K = 4$. Let $L = K(\alpha)$, where $\alpha = \sqrt{2\sqrt{2} - 1}$.

Prove that L is the Hilbert class field of K.

(3) Prove for prime number p, $\exists x, y \in \mathbb{Z}$ s.t. $p = x^2 + 14y^2$ iff $\left(\frac{-14}{p}\right) = 1$ and $(x^2+1)^2 \equiv 8 \pmod{p}$ has an integer root.